Time Series Econometrics

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Exercise 1: Study of `monthly-traffic-fatalities-in-on.txt`

The file `monthly-traffic-fatalities-in-on.txt` contains a time series representing the monthly number of deaths due to fatal road accidents in Ontario between 1960 and 1974. Let us plot its evolution over time:

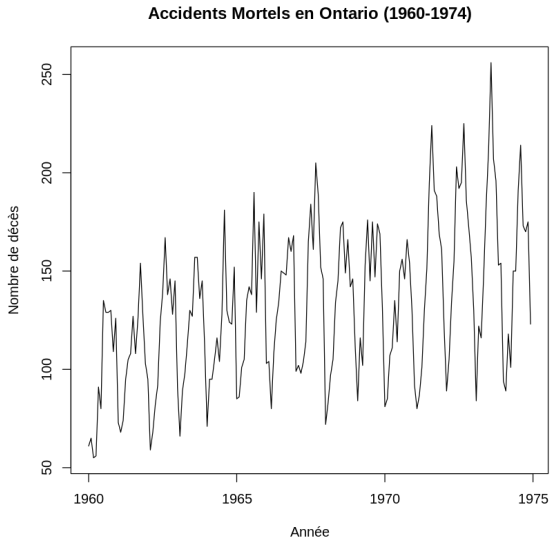

Figure 1: Monthly number of deaths from fatal road accidents in Ontario between 1960 and 1974. We verified that the length of the time series is 180, confirming that all data were correctly loaded from the file.

The time series does not appear to be stationary, as it exhibits an upward trend, although it has a constant variance given the constant width of the curve's envelope. Furthermore, it appears to be seasonal with an annual periodicity. Indeed, over a one-year period, a recurring pattern can be observed suggesting annual seasonality. Let us examine these characteristics in more detail using the autocorrelation coefficients:

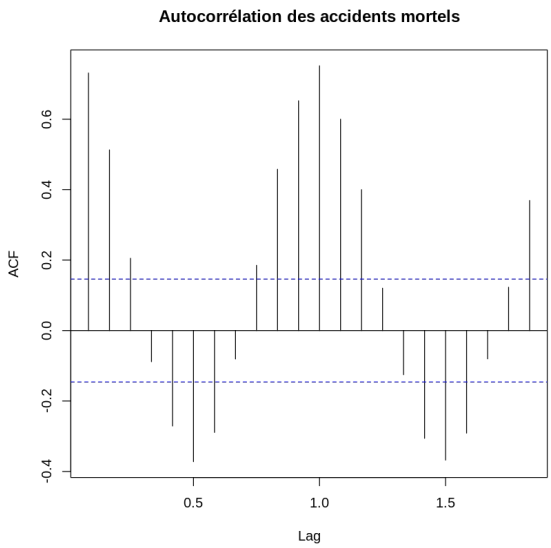

Figure 2: Autocorrelation of fatal accidents.

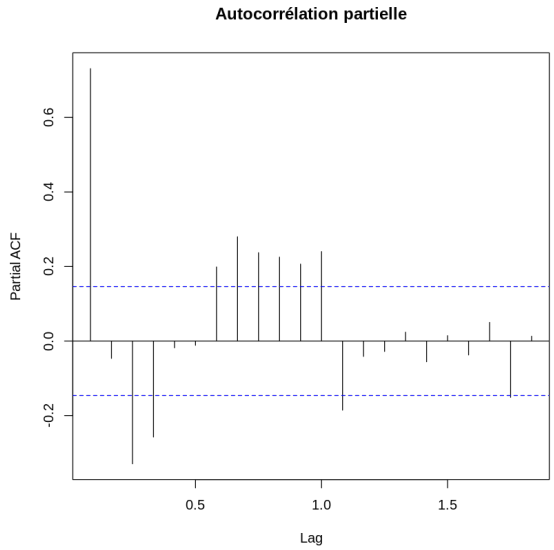

Figure 3: Partial autocorrelation of fatal accidents.

By analysing the ACF, we confirm the presence of annual periodicity: the autocorrelation coefficients repeat every year. Moreover, the series does not appear to be stationary, as the coefficients do not vanish as the lag increases. As for the PACF, it does not reveal a clear structure, since the series is not stationary. There is therefore nothing to conclude about the orders of a potential ARMA model.

Let $k = 1970$. We will produce two forecasts for the twelve months of year $k + 1 = 1971$, using a training set consisting of data from 1960 to $k = 1970$ inclusive — one with parameters $(\alpha, \beta) = (0.2, 0.2)$ and the other with $(\alpha, \beta) = (0.9, 0.9)$. In R, we will not specify the value of γ, as R performs a grid search and automatically optimises this value by testing several possibilities. We will simply ask R for the value of γ after the model has been evaluated.

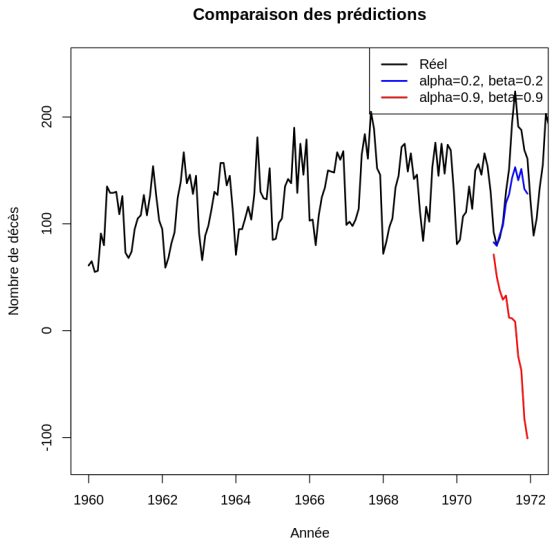

Figure 4: Comparison of forecasts for the monthly number of fatal road accident deaths in Ontario for the year 1971.

We thus obtain two very different models. Let us examine the values of their parameters as well as the mean squared error, computed by summing the mean squared errors of the above forecasts for k ranging from 1963 to 1973.

α	β	γ	Total MSE
0.2	0.2	0.174	6805.569
0.9	0.9	0.574	228692.3

Table 1: Summary table of the Holt-Winters smoothing parameters and their total mean squared errors. The first smoothing model appears to be significantly better than the second.

Finally, we will compute the forecast for the year 1974, based on values from 1960 to 1973, using Holt-Winters smoothing. Once again, we will not specify the parameter values, as R handles their optimisation automatically.

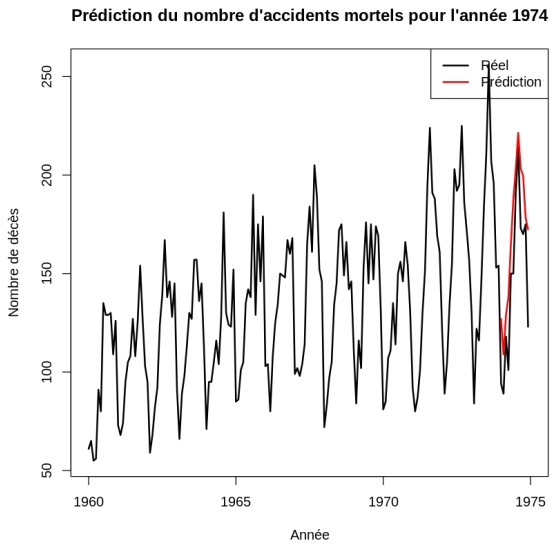

Figure 5: Forecast of the number of fatal accidents for the year 1974 based on values from 1960 to 1973 inclusive. The optimal Holt-Winters smoothing corresponds to the values $\alpha = 0.151$, $\beta = 0.008$, and $\gamma = 0.249$.

Exercise 2: Study of `monthly-australian-imports-from.txt`

The file `monthly-australian-imports-from.txt` contains a time series representing the monthly imports of Japanese goods into Australia in thousands of dollars between July 1965 and October 1993. Let us plot its evolution over time:

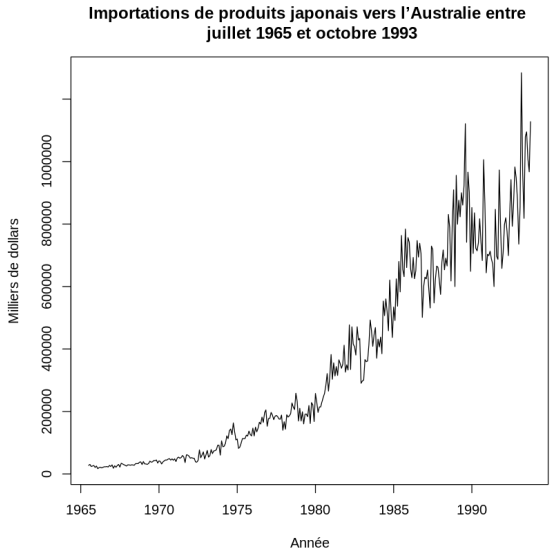

Figure 6: Monthly imports of Japanese goods into Australia in thousands of dollars between July 1965 and October 1993. We verified that the length of the time series is 340, confirming that all data were correctly loaded from the file.

The time series does not appear to be stationary, as it exhibits a strongly upward trend and does not have a constant variance either, given the increasing width of the curve's envelope over time. Let us examine its seasonality, which is difficult to assess by eye, using the autocorrelation coefficients:

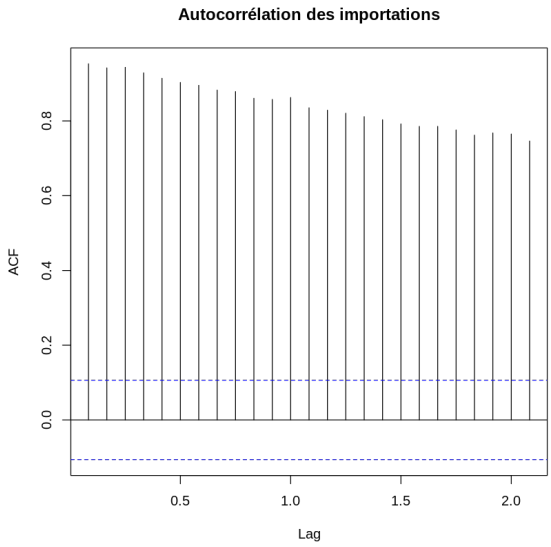

Figure 7: Autocorrelation of imports to Australia.

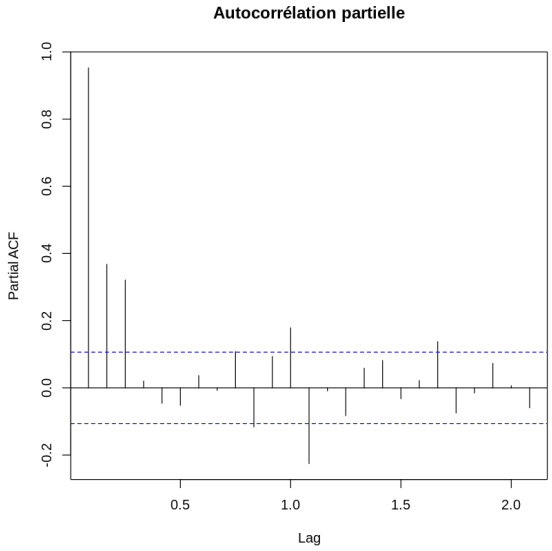

Figure 8: Partial autocorrelation of imports to Australia.

The analysis of the ACF and PACF coefficients shows the absence of periodicity, in contrast to Exercise 1. Examination of the ACF highlights the non-stationarity of the series. Indeed, the envelope of the autocorrelation coefficients decreases very slowly, which reflects a persistent dynamic and energy that does not decay rapidly — the opposite of the behaviour expected for a stationary series. Regarding the PACF, it is not possible to determine the order of a potential ARMA model until stationarity has been established.

In order to continue the study of this series, which we will denote x_t, we must make it stationary. Given the curve in Figure 6, we assume that the series has a polynomial trend of degree 2, that is:

$$x_t = at^2 + bt + c + \epsilon_t \tag{1}$$

where a, b, c are constants and ϵ_t is Gaussian white noise. It follows that:

$$x_{t-1} = a(t-1)^2 + b(t-1) + c + \epsilon_{t-1} \tag{2}$$
$$x_{t-2} = a(t-2)^2 + b(t-2) + c + \epsilon_{t-2} \tag{3}$$

To make x_t stationary, we must eliminate the quadratic and linear components of the trend. To do this, we need to apply the differencing operator Δ twice. Indeed:

$$\begin{aligned} y_t = \Delta^2 x_t &= x_t - 2x_{t-1} + x_{t-2} && (4)\\ &= (at^2 + bt + c + \epsilon_t) - 2(at^2 - 2at + a + bt - b + c + \epsilon_{t-1}) && (5)\\ &\quad + (at^2 - 4at + 4a + bt - 2b + c + \epsilon_{t-2}) && (6)\\ &= 2a + \epsilon_t - 2\epsilon_{t-1} + \epsilon_{t-2} && (7) \end{aligned}$$

The resulting series is therefore an MA(2) process. Let us verify this by examining the autocorrelation coefficients of y_t:

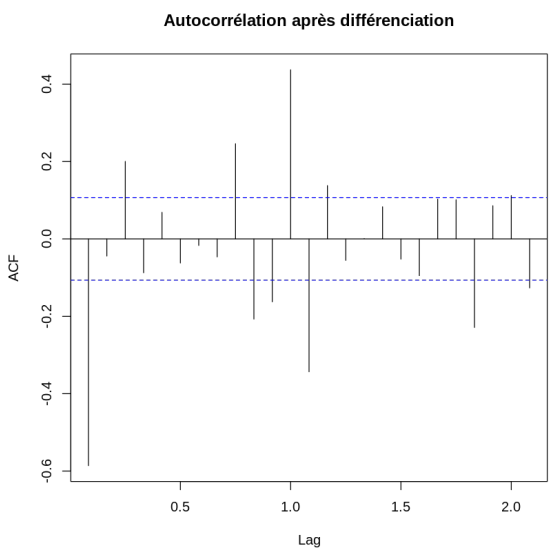

Figure 9: Autocorrelation of $\Delta^2 x_t$.

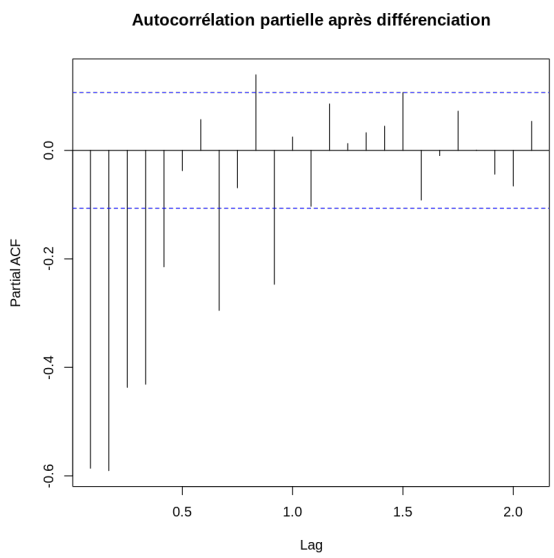

Figure 10: Partial autocorrelation of $\Delta^2 x_t$.

In the ACF, significant spikes are observed at the first lags. The autocorrelation function drops abruptly to zero after lag 2, with subsequent spikes being non-significant as they lie within the interval delimited by the blue dashed lines. This result is consistent with an MA(2) process, since theoretically, for an MA(q) process, the ACF should be zero beyond lag q. As for the PACF, a more gradual decay

is observed. Several significant spikes appear, diminishing progressively in amplitude. This behaviour is characteristic of an MA process.

Let us verify this hypothesis using the `auto.arima` function, which tests several ARMA models and selects the one offering the best fit to the data. The obtained coefficients are as follows:

```
Series: y
ARIMA(5,0,0)(0,0,1)[12] with zero mean

Coefficients:
          ar1      ar2      ar3      ar4      ar5    sma1
      -1.5035  -1.7001  -1.3757  -0.8355  -0.2864  0.4718
s.e.   0.0529   0.0868   0.1019   0.0862   0.0525  0.0553

sigma^2 = 4.089e+09:  log likelihood = -4219.92
AIC=8453.84   AICc=8454.18   BIC=8480.6
```

Figure 11: Coefficients of the ARMA model obtained with the `auto.arima` function.

Applying the `auto.arima` function to our time series suggests an ARIMA(5,0,1) model, which differs significantly from the theoretically expected MA(2). The estimated model presents relatively large autoregressive coefficients (notably $\phi_1 = -1.5035$ and $\phi_2 = -1.7001$) and a large residual variance ($\sigma^2 = 4.089 \times 10^9$). These results, combined with the information criteria (AIC = 8453.84, BIC = 8480.6), suggest that the automatic model identification may not be optimal for this series.

```
        Box-Ljung test

data:  residuals(reg)
X-squared = 40.317, df = 12, p-value = 6.37e-05
```

Figure 12: Box-Ljung test for residual whiteness at level $\alpha = 0.05$.

The Box-Ljung test applied to the residuals of the ARIMA(5,0,1) model yields a test statistic of $\chi^2 = 40.317$ with 12 degrees of freedom and a p-value of 6.37×10^{-5}. This p-value, well below the usual significance threshold of 5%, leads us to reject the null hypothesis of residual whiteness. This indicates that the model does not adequately capture the temporal dependence structure present in the data.

A possible improvement would consist of manually fitting an MA(2) model and comparing the various metrics, such as AIC and BIC, with those of the model selected by `auto.arima`.